

Problem 1

- v. Consider $(x^2 - y, x^3 - z) = A$. We can see that $A \subset \ker$ since $f(x^2 - y) = t^2 - t^2 = 0$ and $f(x^3 - z) = 0$

Consider $\mathbb{C}[x, y, z]/A$. Let $g \in \ker$, now in the quotient ring we can rewrite g s.t.:

$$y = x^2, z = x^3$$

This means this new polynomial is equivalent to some $g'(x) = g(x, x^2, x^3) \in \mathbb{C}[x]$.

Since we know that $g(t, t^2, t^3) = 0$ this implies that $g'(t) = g(t, t^2, t^3) = 0$ for all $t \in \mathbb{C}$. Now since g' is a single variable and is zero for all t this implies it is the zero polynomial. Thus, this implies it is in A . Thus $\ker = A$

Problem 2

(b) From (a) we know that $I^e \subseteq I^{ec}$. Thus, we can let $I = I^e$ thus $I^e \subseteq I^{ece}$

Let $x \in I^{ece} \implies x = \sum_{i=1}^n r_i x_i$ s.t. $r_i \in B$ and $x_i \in \phi(I^{ec})$. This implies there is a $a_i \in I^{ec}$ s.t. $x_i = \phi(a_i)$.

$$x = \sum_{i=1}^n r_i \phi(a_i)$$

Since $a_i \in I^{ec} = \phi^{-1}(I^e)$ this means that $\phi(a_i) \in I^e$ this implies that $x \in I^e$ since ideals are closed under multiplication from any element and addition withing the ideal. Thus $I^e = I^{ece}$

From (a) we know that $J^{ec} \subseteq J$. Thus, we can let $J = J^c \implies J^{cec} \subseteq J^c$

Let $x \in J^{cec} \implies x \in \phi^{-1}(J^{ce}) \implies \phi(x) \in J^{ce}$. Thus, $\phi(x)$ is in the form:

$$\phi(x) = \sum_{i=1}^n r_i x_i$$

s.t. $r_i \in B$ and $x_i \in \phi(J^c) \implies \exists a_i \in J^c = \phi^{-1}(J)$ s.t. $\phi(a_i) = x_i \implies x_i \in J \implies \phi(x) \in J \implies x \in J^c$

Thus, $J^{cec} = J^c$.

(c) Let q be a prime ideal in B . The consider:

$$q^c = \phi^{-1}(q) = \{a \in A \mid \phi(a) \in q\}$$

Let $a, b \in q^c$ this means that $\phi(ab) = \phi(a)\phi(b) \in q$. This means that:

$$\phi(a) \in q \quad \text{or} \quad \phi(b) \in q$$

This implies that

$$a \in q^c \quad \text{or} \quad b \in q^c$$

Consider the natural injective map $\phi: \mathbb{Z} \rightarrow \mathbb{Q}$. Consider the $(0) \subseteq \mathbb{Q}$. This is maximal since $\mathbb{Q}$ is a field, so there is only ideals $(0), \mathbb{Q}$

Consider $(0)^c = \{a \in \mathbb{Z} \mid \phi(a) = 0\} = (0)$ since ϕ is injective. However, $(0) \subset (2)$ is not maximal in $\mathbb{Z}$.

Problem 3

Consider $\varphi : \mathbb{Z}[i]/(5) \rightarrow \mathbb{F}_5 \times \mathbb{F}_5$ s.t. $\varphi(a + bi) = (a - 2b, a + 2b) \pmod{5}$

Injective:

Let $a + bi \in \ker \varphi$ then $\varphi(a + bi) = (0, 0)$. This means that $a = 2b$ and $a = -2b$ this implies $4b = 0$ thus multiply by 4, and we get $b = 0$ thus $a = 0$. Thus, injective.

Surjective:

Let $(c, d) \in \mathbb{F}_5 \times \mathbb{F}_5$. Let $a = 3(c + d) \pmod{5}$ and let $b = 4(d - c) \pmod{5}$. Thus consider $\varphi(a, b) = (a - 2b, a + 2b) \pmod{5} = (3c + 3d - 8d + 8c, 3c + 3d + 8d - 8c) \pmod{5} = (c, d) \pmod{5}$ Thus, surjective.

Homomorphism:

$$\varphi(1) = (1, 1)$$

Let $a + bi, c + di \in \mathbb{Z}[i]/(5)$

$$\varphi(a + c + (b + d)i) = (a + c - 2b - 2d, a + c + 2b + 2d) = (a - 2b, a + 2b) + (c - 2d, c + 2d) = \varphi(a + bi) + \varphi(c + di)$$

$$\varphi((a + bi)(c + di)) = \varphi(ac + adi + cbi - bd) = \varphi(ac - bd + (ad + cb)i) = (ac + 4bd - 2ad - 2cb, ac + 4bd + 2ad + 2cb) = ((a - 2b)(c - 2d), (a + 2b)(c + 2d)) = \varphi(a + bi)\varphi(c + di)$$

Thus $\mathbb{Z}[i]/(5) \cong \mathbb{F}_5 \times \mathbb{F}_5$

Proven in HW1 we know that ideals of $\mathbb{F}_5 \times \mathbb{F}_5$ are of the form ideals in $I \times J$ s.t. I is an ideal in $\mathbb{F}_5$. Thus, since $\mathbb{F}_5$ is a field we know the only ideals are $(0), \mathbb{F}_5$. Thus, we have 4 ideals:

- $(0) \times (0)$
- $\mathbb{F}_5 \times \mathbb{F}_5$
- $\mathbb{F}_5 \times (0)$
- $(0) \times \mathbb{F}_5$

Mapping these back to $\mathbb{Z}[i]/(5)$ we get $(0) \times (0) \cong (0), (0) \times \mathbb{F}_5 \cong (2 - i), \mathbb{F}_5 \times (0) \cong (2 + i), \mathbb{F}_5 \times \mathbb{F}_5 \cong (1)$.

(0) is not prime since $(2 - i)(2 + i) = 5 = 0$. Thus, not maximal.

(1) is not prime and maximal by definition.

$(0) \times \mathbb{F}_5$ is maximal since if we add anything to the (0) part since $\mathbb{F}_5$ is a field everything is a unit except zero, so we would get the whole ring. Thus, $(2 - i), (2 + i)$ are maximal thus prime.

Problem 4

(a) iv. Consider $7 = (4x + 5)(4x - 5) - 8(2x^2 - 4)$. Thus, $7 \in (2x^2 - 4, 4x - 5)$.

$$4x - 5 \equiv 0 \pmod{7}$$

$$4x \equiv 5 \pmod{7}$$

$$x \equiv 3 \pmod{7}$$

Thus $(2x^2 - 4, 4x - 5) = (x - 3, 7)$ since $4(x - 3) + 7 = 4x - 5$ and $(2x + 6)(x - 3) + 14 = 2x^2 - 4$. Thus:

$$\frac{\mathbb{Z}[x]/(x - 3)}{(x - 3, 7)/(x - 3)} \cong \mathbb{Z}/(7) \cong \mathbb{Z}/7\mathbb{Z}$$

(b) Assume for contradiction they are isomorphic.

This implies that $\mathbb{Z}[x]/(x^2 + 7, 2) \cong \mathbb{Z}[x]/(2x^2 + 7, 2)$.

However, this implies that $\mathbb{Z}[x]/(2x^2 + 7, 2) = \mathbb{Z}[x]/(0 + 7, 2) = (0)$ since $7 - 2(3) = 1$.

However, $\mathbb{Z}[x]/(x^2 + 7, 2) = \mathbb{Z}[x]/(x^2 + 1, 2) \cong \mathbb{F}_2[x]/(x^2 + 1) = \{0, 1, x, 1 + x\} \neq (0)$.

(c) If $p = 2$:

$$\mathbb{Z}[i]/(2) \cong \frac{\mathbb{Z}[x]/(x^2 + 1)}{(2)} \cong \frac{\mathbb{Z}[x]/(x^2 + 1)}{(2, x^2 + 1)/(x^2 + 1)} \cong \mathbb{Z}[x]/(2, x^2 + 1)$$

$$\mathbb{Z}[x]/(2, x^2 + 1) \cong \frac{\mathbb{Z}[x]/(2)}{(2, x^2 + 1)/(2)} \cong \mathbb{F}_2[x]/(x^2 + 1) = \mathbb{F}_2[x]/(x - 1)^2$$

Let

$$R_1 = \mathbb{F}_2[x]/(x - 1)^2,$$

$$R_2 = \mathbb{F}_2[x]/(x^2).$$

Define a map:

$$\varphi : \mathbb{F}_2[x] \rightarrow \mathbb{F}_2[x]$$

by:

$$\varphi(f(x)) = f(x + 1)$$

This is an automorphism. Define a map:

$$\psi : \mathbb{F}_2[x] / ((x-1)^2) \rightarrow \mathbb{F}_2[x] / (x^2)$$

by:

$$\psi(a + ((x-1)^2)) = \varphi(a) + \varphi((x-1)^2) = \varphi(a) + (x^2)$$

Well-Defined:

Assume that $r + ((x-1)^2) = s + ((x-1)^2)$ for some $r, s \in \mathbb{F}_2[x]$. Then

$$r - s \in ((x-1)^2).$$

Since φ is a ring homomorphism, we have

$$\varphi(r - s) = \varphi(r) - \varphi(s) \in (x^2).$$

Thus,

$$\varphi(r) + (x^2) = \varphi(s) + (x^2),$$

which shows that $\psi(r + ((x-1)^2)) = \psi(s + ((x-1)^2))$. Therefore, the map is well-defined.

Homomorphism:

For $r, s \in \mathbb{F}_2[x]$, we have

$$\psi((r+s) + ((x-1)^2)) = \varphi(r+s) + (x^2) = (\varphi(r) + \varphi(s)) + (x^2),$$

$$\psi(rs + ((x-1)^2)) = \varphi(rs) + (x^2) = \varphi(r)\varphi(s) + (x^2),$$

Bijjective

Since φ is an automorphism of $\mathbb{F}_2[x]$, it is bijective. Its inverse φ^{-1} is also an automorphism. The inverse of ψ is given by

$$\psi^{-1} : R/(x^2) \rightarrow R/((x-1)^2), \quad \psi^{-1}(r + (x^2)) = \varphi^{-1}(r) + ((x-1)^2).$$

Thus, ψ is bijective.

Conclusion.

Since ψ is a well-defined ring homomorphism that is bijective, we conclude that

$$\mathbb{F}_2[x] / ((x-1)^2) \cong \mathbb{F}_2[x] / (x^2).$$

Let p be odd and assume there exists a $a \in \mathbb{Z}$ s.t. $a^2 \equiv -1 \pmod{p}$

$$\begin{aligned}\mathbb{Z}[i]/(p) &\cong \frac{\mathbb{Z}[x]/(x^2+1)}{(p)} \cong \frac{\mathbb{Z}[x]/(x^2+1)}{(p, x^2+1)/(x^2+1)} \cong \mathbb{Z}[x]/(p, x^2+1) \\ \mathbb{Z}[x]/(p, x^2+1) &\cong \frac{\mathbb{Z}[x]/(p)}{(p, x^2+1)/(p)} \cong \mathbb{F}_p[x]/(x^2+1) \cong \mathbb{F}_p[x]/(x-a)(x+a)\end{aligned}$$

Thus, since $x-a-x-a = -2a$. Since p is odd this implies that $-2a \neq 0$. Since we are in $\mathbb{F}_p$ we know we can find an inverse for $-2a$ let that be b . Thus, $b(x-a) + b(x+a) = b(-2a) = 1$ Thus by the CRT:

$$\mathbb{F}_p[x]/(x-a)(x+a) \cong \mathbb{F}_p[x]/(x-a) \times \mathbb{F}_p[x]/(x+a)$$

Define $\varphi : \mathbb{F}_p[x] \rightarrow \mathbb{F}_p$ s.t. $f(x) \mapsto f(a)$

Homomorphism:

- $\varphi(f(x) + g(x)) = f(a) + g(a) = \varphi(f(x)) + \varphi(g(x))$
- $\varphi(f(x)g(x)) = f(a)g(a) = \varphi(f(x))\varphi(g(x))$
- $\varphi(1) = 1$

This is clearly surjective since $f(k) = k$ for $k \in \mathbb{F}_p$

The kernel is also all polynomial's with a root of a thus this is the set divisible by $x-a$ which his $(x-a)$

Thus by the first isomorphism theorem $\mathbb{F}_p[x]/(x-a) \cong \mathbb{F}_p$ do the same for $\mathbb{F}_p[x]/(x+a) \cong \mathbb{F}_p$

$$\mathbb{F}_p[x]/(x-a) \times \mathbb{F}_p[x]/(x+a) \cong \mathbb{F}_p \times \mathbb{F}_p$$

Let p be odd s.t. there is no integer s.t. $x^2 = -1 \pmod{p}$

Ok do the same thing as above, and we get to $\mathbb{Z}[i]/(p) \cong \mathbb{F}_p[x]/(x^2+1)$. Since there is no integer s.t. $x^2 = -1$ this implies that (x^2+1) is irreducible in $\mathbb{F}_p[x]$. Thus proven in class since this is not zero we know that (x^2+1) is prime and maximal. Thus proven in class since (x^2+1) is maximal this implies that $\mathbb{F}_p[x]/(x^2+1)$ is a field.

Now for the size. Since we have the relation $x^2 = -1$ this implies that we can write every polynomial in $\mathbb{F}_p[x]$ as:

$$a + bx, \quad a, b \in \mathbb{F}_p$$

The reason is if we have a term cx^n s.t. $c \in \mathbb{F}_p$ and $n > 1$. We can use the division algorithm and get $n = 2q + r$ s.t. $q \in \mathbb{N}$ and $r \in \{0, 1\}$

$$\text{Thus } cx^n = c(-1)^q x^r$$

Thus, since a, b are chosen independently in $\mathbb{F}_p$ we get p choices for a and p for b thus we get p^2 elements.

(d)

$$\mathbb{Z}[\omega]/(p) \cong \begin{cases} \mathbb{F}_3[t]/(t^2), & \text{if } p = 3, \\ \mathbb{F}_p \times \mathbb{F}_p, & \text{if } p \neq 3 \text{ and } \exists a \in \mathbb{Z} \text{ s.t. } a^3 = 1, a \neq 1 \pmod{p}, \\ \text{a field with } p^2 \text{ elements,} & \text{if } p \neq 3 \text{ and there is not an integer as described above.} \end{cases}$$

Define $\varphi : \mathbb{Z}[\omega] \rightarrow \mathbb{Z}[x]$ s.t. $\varphi(f(x)) = f(\omega)$ This is the same proof as in (c). This is a surjective homomorphism with kernel $(x^2 + x + 1)$ thus $\mathbb{Z}[\omega] \cong \mathbb{Z}[x]/(x^2 + x + 1)$

If $p = 3$

$$\begin{aligned} \mathbb{Z}[\omega]/(3) &\cong \frac{\mathbb{Z}[x]/(x^2 + x + 1)}{(3)} \cong \frac{\mathbb{Z}[x]/(x^2 + x + 1)}{(3, x^2 + x + 1)/(x^2 + x + 1)} \cong \mathbb{Z}[x]/(3, x^2 + x + 1) \\ \mathbb{Z}[x]/(3, x^2 + x + 1) &\cong \frac{\mathbb{Z}[x]/(3)}{(3, x^2 + x + 1)/(3)} \cong \mathbb{F}_3[x]/(x^2 + x + 1) = \mathbb{F}_3[x]/(x - 1)^2 \end{aligned}$$

Use a similar proof for the $p = 2$ case in part (c) to show that $\mathbb{F}_3[x]/(x - 1)^2 \cong \mathbb{F}_2[x]/(x^2)$

If $p \neq 3$ and there is an $a \in \mathbb{Z} \setminus \{1\}$ s.t. $a^3 = 1, a \neq 1 \pmod{p}$.

Then again we get $\mathbb{Z}[\omega]/(p) \cong \mathbb{F}_p[x]/(x^2 + x + 1)$. Since $a^3 = 1 \implies a^3 - 1 = 0 \implies (a - 1)(a^2 + a + 1) = 0$ and since $a \neq 1$ we know that $a - 1$ is not zero thus we can cancel it since we are in a field, and we get that $(a^2 + a + 1) = 0$. Now consider $(x - a)(x + a + 1) = x^2 + ax + x - ax - a^2 - a$. Since $a^2 + a = -1$, $x^2 + ax + x - ax - a^2 - a = x^2 + x + 1$.

Since $x + a + 1 - x + a = 2a + 1$. We get that these are coprime since we can get to one since we are in a field. Thus by CRT

$$\mathbb{F}_p[x]/(x - a)(x + a + 1) \cong \mathbb{F}_p[x]/(x - a) \times \mathbb{F}_p[x]/(x + a + 1)$$

Now do a similar proof from part (c) and we get $\cong \mathbb{F}_p \times \mathbb{F}_p$

If $p \neq 3$ and we do not have that property then this means that $(x^2 + x + 1)$ is irreducible thus we do the same proof to show that $\mathbb{Z}[\omega]/(p)$ is a field with p^2 elements.

Problem 5

(a) Let $E = \{0\}$. Since every ideal has 0 this implies that $V(0) = \text{Spec}(A)$. Thus, X is closed.

Let $E = \{1\}$. Since if an ideal contains 1 then it is the whole ring, and since prime ideals are not the whole ring we get that $V(1) = \emptyset$. Thus, $\emptyset$ is closed.

Let $\{E_i\}_{i \in I}$ be a collection of subsets of A . Consider:

$$E = \bigcap_{i \in I} V(E_i)$$

By definition a prime ideal p belongs to $V(E_i) \iff E_i \subseteq p$. Thus if $p \in E$ then $E_i \subseteq p$ for all $i \in I$. Thus:

$$\left(\bigcup_{i \in I} E_i \right) \subseteq p$$

Thus:

$$p \in V\left(\bigcup_{i \in I} E_i\right) = B$$

Let $p \in B$ thus

$$\left(\bigcup_{i \in I} E_i \right) \subseteq p$$

Thus we know that $E_i \subseteq p$ for all $i \in I$. Thus, we know that $p \in E$.

Thus, $E = B$, E is closed since B is closed.

Consider $V(E) \cup V(F)$ for $E, F \subseteq A$. Consider EF the ideal generated by products of ef s.t. $e \in E$ and $f \in F$

Let $p \in V(E) \cup V(F)$. Then either $E \subseteq p$ or $F \subseteq p$. In either case, if $e \in E$ and $f \in F$, then either $e \in p$ or $f \in p$. Since p is a prime ideal, it follows that $ef \in p$. Thus, every element of EF is in p , and so $p \in V(EF)$.

Let $p \in V(EF)$ thus $EF \subseteq p$. Assume for contradiction that $p \notin V(E)$ and $p \notin V(F)$. Then there exists $e \in E$ and $f \in F$ s.t. $e \notin p$ and $f \notin p$. But since p is prime, this means that $ef \notin p$. This is a contradiction. Thus either $E \subseteq p$ or $F \subseteq p$ thus $p \in V(E) \cup V(F)$.

Thus, $V(E)$ satisfy the axioms of closed sets.



